STANDARD OPERATING PROCEDURE

L1 Network Connectivity Troubleshooting via Command Prompt

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Department	IT Service Desk / L1 Tech Support
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Owner	L1 Helpdesk Team Lead

1. Objective

To provide L1 Helpdesk technicians with an explicit, repeatable, step-by-step procedure to diagnose, isolate, and resolve a user's inability to connect to the internet or internal company network resources using native Windows command-line tools. Every step below states exactly what to type, what result to expect, and what to do next based on that result.

2. Scope

This SOP applies to all corporate Windows workstations (Windows 10/11) managed by the L1 Helpdesk. It covers wired and Wi-Fi connectivity issues where the OS boots normally and the user can reach a command prompt. It does not cover hardware replacement, VPN client issues, or server-side/domain controller diagnostics — see SOP-NET-002 and SOP-NET-003 for those.

3. Roles & Responsibilities

- L1 Technician: Executes Steps 1–7 below, documents results in the ticket, and escalates per Section 7 if the issue is not resolved.
- L2 Network Team: Receives escalated tickets with full diagnostic data attached; investigates switch ports, DHCP scope, and firewall/routing issues.
- End User: Provides remote access/physical presence at the machine and confirms when connectivity is restored.

4. Pre-requisites & Warnings

⚠️ **Local Administrator privileges are required on the target machine before starting Step 3.**
⚠️ Warn the user before Step 4 that their network connection will drop for a few seconds during the release/renew process.
⚠️ Do not proceed past Step 2 if the user's account is locked out or the machine cannot boot — escalate directly to L2 instead.

5. Step-by-Step Procedure

Step 1: Open Command Prompt with Elevated Privileges

What to do:

- Press the Windows key, type cmd, and wait for "Command Prompt" to appear in the search results.
- Right-click Command Prompt and select "Run as administrator."
- Click Yes on the User Account Control (UAC) prompt.

Expected result: A black terminal window opens with the title bar reading "Administrator: Command Prompt."

If the UAC prompt does not appear or Yes is greyed out: the user's account lacks admin rights. Stop here and escalate to L2 for a temporary admin grant. Do not continue to Step 2.

Step 2: Verify Local IP Configuration

What to do: run the following command exactly as shown:

```
ipconfig /all
```

Locate the active network adapter in the output (labelled "Ethernet adapter Ethernet" for wired, or "Wireless LAN adapter Wi-Fi" for wireless) and record the following three values from the ticket:

- IPv4 Address
- Subnet Mask
- Default Gateway

Decision point:
• If the IPv4 Address starts with 169.254.x.x → the adapter failed to obtain a DHCP lease. Go to Step 3.
• If the IPv4 Address is a normal address (e.g. 10.x.x.x or 192.168.x.x) and a Default Gateway is listed → DHCP is working. Skip Step 3 and go directly to Step 4.
• If no adapter is listed as "Connected" at all → this is a physical/driver issue, not a software one. Check the cable/Wi-Fi toggle, then escalate to L2 hardware if the adapter still does not show as connected.

Step 3: Release, Renew, and Flush Network Settings

What to do: run each command in order, waiting for each to finish before running the next:

```
ipconfig /release
ipconfig /renew
ipconfig /flushdns
```

Expected result: "ipconfig /renew" returns a new IPv4 Address that does NOT start with 169.254, and "ipconfig /flushdns" returns "Successfully flushed the DNS Resolver Cache."

If the address after /renew still starts with 169.254.x.x → the machine cannot reach a DHCP server. Do not repeat this step more than twice. Escalate to L2 with the ipconfig /all output attached (see Section 7).

Step 4: Run Layered Connectivity Tests (Ping Verification)

What to do: run each ping test in order. Let each one complete (4 replies or timeouts) before moving to the next:

4a. Test the local TCP/IP stack:

```
ping 127.0.0.1
```

Expected: 4 replies from 127.0.0.1. If this fails, the machine's TCP/IP stack itself is broken — escalate to L2 immediately; do not continue.

4b. Test the local gateway (use the Default Gateway address recorded in Step 2):

```
ping [Default_Gateway_IP]
  e.g.  ping 192.168.1.1
```

Expected: 4 replies with reasonable latency (under ~50 ms on a LAN).

4c. Test external public internet connectivity:

```
ping 8.8.8.8
```

Expected: 4 replies from 8.8.8.8.

Decision point — read carefully, this determines where the fault lies:
• 4a fails → local stack broken. Escalate to L2 (possible OS/driver corruption).
• 4a succeeds, 4b fails → the machine cannot reach its own gateway. Check physical cable/Wi-Fi signal first; if confirmed connected physically, escalate to L2 (switch port or local network issue).
• 4a and 4b succeed, 4c fails → local network is fine but external routing is broken. Go to Step 5 (trace the path).
• All three succeed → basic IP connectivity is restored. Go to Step 6 to confirm DNS/name resolution before closing the ticket.

Step 5: Trace the Network Path (Path Isolation)

Only perform this step if Step 4c (ping 8.8.8.8) failed while 4a and 4b succeeded.

What to do:

```
tracert 8.8.8.8
```

Read the output line by line, top to bottom. Each line is one "hop" the packet passes through.

Expected result: Hops proceed with reply times and no more than 1-2 consecutive "Request timed out" lines before reaching the destination.

Decision point:
• If the trace stops responding (Request timed out on every hop) starting at hop 1 or 2 → the fault is at the local gateway/router. Escalate to L2.
• If the trace completes several hops successfully then times out consistently from a hop onward → the fault is further upstream (ISP/WAN). Escalate to L2 with the full tracert output attached — do not attempt further local troubleshooting.

Step 6: Confirm DNS Name Resolution

Only perform this step if Step 4 fully succeeded (all three pings replied).

What to do:

```
nslookup www.google.com
```

Expected result: The command returns a server name and one or more IP addresses ("Non-authoritative answer") without an error.

• If nslookup returns "DNS request timed out" or "Can't find www.google.com" while the ping tests in Step 4 succeeded → this is a DNS-specific issue, not general connectivity. Re-run ipconfig /flushdns (Step 3) once; if it persists, escalate to L2 and note "DNS resolution failure" specifically in the ticket.

Step 7: Confirm Resolution With the User

- Ask the user to open a web browser and confirm a normal website (e.g. company intranet) loads.
- Ask the user to confirm they can access any specific internal resource they originally reported as broken (shared drive, printer, application, etc.).

Expected result: User confirms normal access is restored. Proceed to Section 6 to close the ticket.

6. Expected Outcome

The network adapter obtains a valid, non-169.254 IP address; ping tests to loopback, gateway, and 8.8.8.8 all succeed; DNS resolution succeeds; and the user confirms normal access is restored. If any decision point above routes to escalation instead, this SOP's job is complete once the diagnostic data has been captured and handed to L2 per Section 7.

7. Escalation & Documentation

Before closing OR escalating any ticket under this SOP, the technician must:

- Copy and paste the full output of ipconfig /all, the three ping tests, and (if run) tracert and nslookup into the ticket notes.
- Record which Step the issue was resolved at, or which decision point triggered escalation.

Escalate to the L2 Network Team when any of the following are true:

- The local TCP/IP stack test (Step 4a, ping 127.0.0.1) fails.
- The machine cannot reach its own default gateway (Step 4b) after confirming physical connectivity.
- DHCP renewal (Step 3) still returns a 169.254.x.x address after two attempts.
- tracert shows the path failing at the gateway or upstream, per Step 5.
- Total troubleshooting time on this SOP exceeds 20 minutes without resolution.

Escalation ticket must include: user name, asset tag/hostname, all captured command outputs, and the specific decision point reached.

8. Revision History

Version	Date	Author	Description of Change
1.0	July 2026	IT Service Desk	Initial release
2.0	July 28, 2026	IT Service Desk	Rewritten with explicit commands, expected outputs, decision branches, and defined escalation criteria